



**Faculty of Computers &
Artificial Intelligence**



VALEO COMPANY



Benha University

Driver Monitoring System (DMS)

Graduation Project

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Abstract

It goes without saying that we extremely became dependent on means of transportation, especially cars, so safety is a must for drivers, specifically for those who drive for a long time.

When it comes to safety, it must be mentioned that in order for a driver to be fit to drive, he must be focused, driving with full concentration and alert. But first, what is the meaning of drowsiness? Drowsiness is the feeling heavy-eyed or exhausted and being unable to keep your eyes open. The drowsiness levels differ between drivers, which means that the risk levels also differ on the road.

By reviewing other related work about this problem, we found the following techniques used: monitoring steering wheel movement as when driver doesn't have full control on the steering wheel it will lead to unusual driving pattern, tracking the acceleration and the brakes the user makes during his ride using sensors, and the last technique is Computer Vision techniques and deep learning for driver monitoring: this is done by detecting the driver's face using a camera and process the input using deep learning to output his drowsiness level based on yawning or blinking rate. After reviewing these techniques, we found the last technique the most efficient and suitable.

Our project aims to develop a system for detecting driver's drowsiness such as yawning or blinking and taking action based on driver's drowsiness level (low alert for low level ,high alert for high level and call to an emergency contact) using deep learning which will get the picture frames from the camera present in the car and will use jetson Nano as a hardware to process the model and output the result of the drowsiness and output an alert which will then lead to a safer driving for long hours without fear of falling asleep.

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